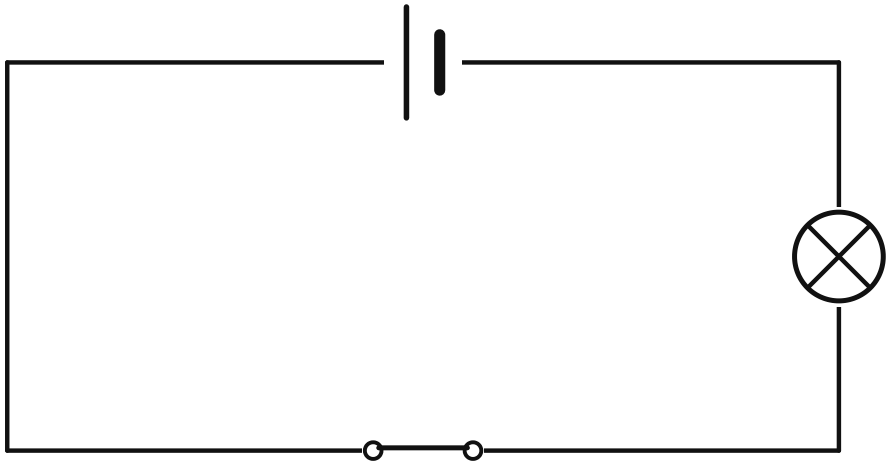
PHYSICAL AND CHEMICAL SCIENCE

Electricity

Things that use electricity, safety, simple circuits and switches, and where electricity comes from



In this pack

- **Learn it:** Electricity and circuits ★☆☆
- **Fact check:** Circuit facts ★☆☆
- **Think it through:** Safe or not safe? ★★☆
- **Investigate:** Make a circuit ★★☆
- **Real-world case study:** Light from the sun ★★★
- **Review and challenge:** Show what you know ★★★

This pack belongs to: _____ **Class:** _____

© InquiryMind · Learn · Play · Wonder · Free resources for everyone, bringing quality and equity to education and helping teachers build concepts and make learning visible · Illustrations drawn for InquiryMind · Pictures from OpenMoji (openmoji.org), CC BY-SA 4.0 · Print in black and white on A4 · Challenge: ★ core ★★ deeper ★★★ stretch

LEARN IT · C4 · ELECTRICITY

Electricity and circuits

★☆☆ Grade 1

Name: Class: Date:

Learning goals

- Name the parts of a circuit.
- Explain what a switch does.

Lights, fans, phones and televisions use **electricity**. Some things get electricity from the **wall socket**. Some use **batteries** (cells).



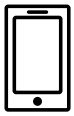
lamp



fan



TV



phone



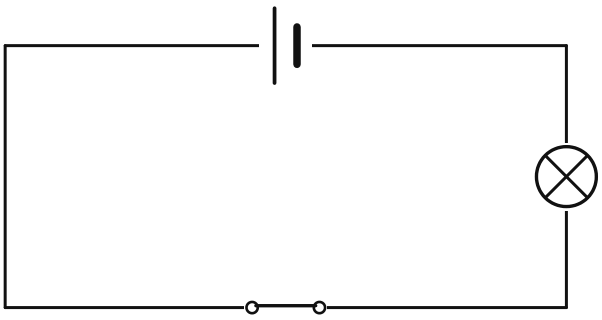
torch

Stay safe

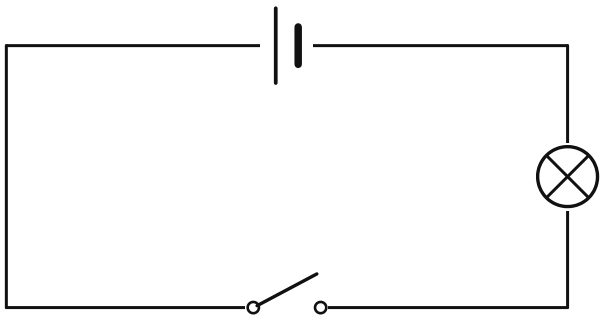
Electricity from the wall is **dangerous**. Never touch sockets or wires, and keep water away from them.

A simple circuit

A **circuit** is a loop. It has a **cell**, **wires** and a **lamp**. Electricity flows round the loop from the cell, through the lamp and back. The loop must be **complete**, with no gaps.



Switch on: the lamp lights.



Switch off: a gap, no light.

A **switch** makes a gap in the circuit (off) or closes the gap (on).

Where does electricity come from?



sunlight (solar panels)



wind



moving water



burning gas or coal

Key words

circuit *a* complete loop for electricity

cell *a* battery that stores electricity

switch turns *a* circuit on *and* off

FACT CHECK · C4 · ELECTRICITY

Circuit facts

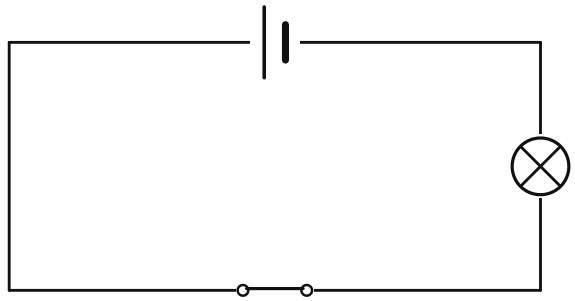
★☆☆
Grade 1

Name: **Class:** **Date:**

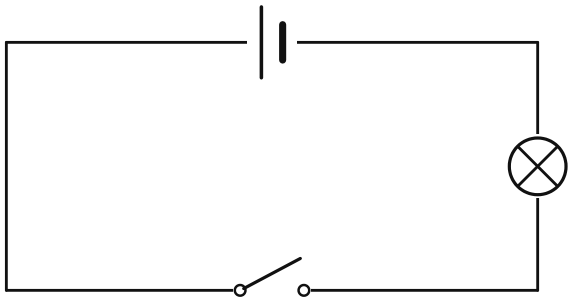
Learning goals

- Name circuit parts.
- Spot complete circuits.

1 Will the lamp light? Write yes or no.



A



B

A	B

2 Match the part to its job.

Draw a line to match each pair, or write the letter.

cell ●	● **A** pushes electricity round
wire ●	● **B** turns it on and off
switch ●	● **C** carries electricity
lamp ●	● **D** gives light

THINK IT THROUGH · C4 · ELECTRICITY

Safe or not safe?

★★☆
Grade 1

Name: Class: Date:

Learning goals

- Think about electrical safety.

1 Sort these.

SORT THESE

using a torch with batteries **asking an adult to plug in**
touching a switch with wet hands **poking a socket**

Safe	Not safe

2 Why does the lamp go out when you open the switch?

INVESTIGATE · C4 · ELECTRICITY

Make a circuit

★★☆
Grade 1

Name: Class: Date:

Learning goals

- Build a circuit.
- Find out what makes it work.

You will need

a cell (battery), two wires with clips, a small bulb in a holder, a switch.

Join the cell, wires and bulb in a loop. Then add the switch.

1 Draw your circuit.

My circuit

2 What happened when you took one wire off?

REAL-WORLD CASE STUDY · C4 · ELECTRICITY

Light from the sun

★★★
Grade 1

Name: Class: Date:

Learning goals

- Read about solar electricity.

Shuvo lives on a char, an island in a big river. There are no electricity wires there.

His family has a **solar panel** on the roof. In the day it turns sunlight into electricity and stores it in a battery. At night the battery lights two lamps, so Shuvo can do his homework.

1 **Where does Shuvo's electricity come from?**

◯ the Sun ◯ a wall socket far away

2 **Why is a solar panel useful on the char?**

REVIEW AND CHALLENGE · C4 · ELECTRICITY

Show what you know

★★★
Grade 1

Name: **Class:** **Date:**

Learning goals

- Use electricity words.

1 Find the words.

E	G	L	A	M	P	C
P	P	I	B	A	G	H
L	N	U	D	B	B	S
S	O	C	E	L	L	W
U	L	O	I	L	G	I
N	V	T	P	S	I	R
K	F	N	L	I	U	E

cell lamp loop sun wire

2 A lamp lights when the circuit is...

◯ complete ◯ broken

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
name the parts of a circuit	◯	◯	◯
make a lamp light	◯	◯	◯
stay safe with electricity	◯	◯	◯